

PARAG PAWAR

AI Engineer | ML Engineering | RAG & LLM Systems

Pune, Maharashtra, India | paragpawar0077@gmail.com | +91-9529747942 | [LinkedIn](#) | [GitHub](#)

SUMMARY

Comfortable across the full pipeline — data cleaning, feature engineering, model training, **API-based deployment**, and monitoring — using Python, Scikit-learn, XGBoost, and SQL, with additional experience **deploying** LLM/RAG-based applications **via Flask microservices**. Seeking an **AI Engineer / Data Science internship** to apply ML modeling and **deployment** skills to real-world, production-facing problems.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java
- **ML / AI:** Scikit-learn, XGBoost, Isolation Forest, One-Class SVM, SMOTE, LLMs, RAG, ChromaDB, Whisper
- **Data:** Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Data Cleaning, Statistical Analysis
- **Databases:** PostgreSQL, SQLite, CTEs, Window Functions, Joins, Aggregations
- **Tools & APIs:** Flask, Streamlit, Google Gemini API, GROQ API, Power BI (Power Query, Data Modeling, DAX), Git, GitHub, Jupyter, VS Code

WORK EXPERIENCE

Fraud Detection Engineer (Virtual Externship) | Zetheta Algorithms Pvt. Ltd. | Feb – Mar 2026

[Certificate](#) | [GitHub](#)

- Built a hybrid ensemble scoring system (XGBoost + Isolation Forest + rule-based signals) on 10,000 transactions; threshold sensitivity analysis (0.10–0.90) identified an optimal threshold of 0.75, achieving **96.7% recall** (29/30 frauds detected, 1 missed) with zero false positives.
- Deployed a risk-tier classification system (High/Medium/Low) that caught **100% of frauds (30/30)** at a total operational cost of Rs. 600 (12 false-positive reviews) — a ~96% cost reduction vs. an estimated Rs. 15,000 baseline loss; .built a UUID-based alert pipeline with explainable risk reasons, designed for **continuous threshold re-evaluation as transaction patterns shift**.

PROJECTS

Customer Churn Prediction | Python, Scikit-learn, PostgreSQL, Power BI [GitHub](#)

- Cleaned and engineered features on the Telco dataset (**7,043 customers, 21 features**), creating 4 new features — tenure group, avg. monthly spend, services count, and auto-payment flag.
- Applied SMOTE and StratifiedShuffleSplit to address class imbalance; selected Gradient Boosting for production (**ROC-AUC 0.838**, Churn Recall 0.80) over Logistic Regression and Random Forest.
- Quantified **Rs. 1.39 lakh** in monthly at-risk revenue (26.54% churn rate); found month-to-month contracts churn at 42.71% vs. 2.83% for two-year contracts.
- Found electronic-check and non-autopay customers churn at a significantly higher rate than autopay users; recommended targeted autopay migration incentives to reduce voluntary churn.
- Wrote 10 SQL queries (CTEs, RANK + PARTITION BY) in PostgreSQL and built a Power BI dashboard with 4 DAX measures.

AI Teaching Assistant (RAG System) | Python, Flask, Whisper, ChromaDB, GROQ [GitHub](#)

- **Built a Flask REST API integrating Whisper (speech-to-text, Hindi→English translation) into a production-ready RAG pipeline.**
- Implemented ChromaDB vector search (cosine similarity, HNSW) with a GROQ-hosted LLaMA 3.3 70B model to generate real-time, timestamped answers, reducing hallucinations via prompt engineering.

SQL Query Assistant | Python, Gemini API, SQLite, Streamlit [GitHub](#)

- Built natural-language-to-SQL conversion with dynamic schema injection into the Gemini prompt; handles TOP-N, ranking, and aggregation queries across an 11-table Chinook database.
- Designed a 5-module architecture with caching, session-state query history, and API quota error handling.

Brazilian E-Commerce Sales Analysis | Python, PostgreSQL, Power BI [GitHub](#)

- Analyzed **99,441** real Olist orders across 7 relational tables, cleaning and translating category data into an analysis-ready dataset.
- Wrote 10 SQL queries using multi-table joins, CTEs, LAG(), and EXTRACT+EPOCH date arithmetic to uncover order and delivery trends.
- Built a Power BI dashboard live-connected to PostgreSQL with 6 DAX measures for interactive drill-down analysis.

ACHIEVEMENTS

- Contributed a merged pull request (**#1034**) to Free-TV/IPTV, a public repository with **18.8K+ GitHub stars**, fixing broken stream validation entries to maintain playlist reliability. [PR #1034](#)
- Collaborated as part of Team CODESTROM at VYOMA 2026, a national-level prototype competition hosted by AISSMS IOIT, Pune (Mar 2026). [Certificate](#)

EDUCATION

Bhivarabai Sawant Institute of Technology and Research (BSIOTR), JSPM Pune, India

B.E. in Computer Engineering | SPPU | CGPA: 7.18 — Aug 2023 – May 2027

- Relevant Coursework: DSA, OOP, DBMS, Operating Systems, Computer Networks, Discrete Mathematics